INTERACTIVE EDA DASHBOARD

Final Project Review | Department of Computer Science and Engineering | KL Deemed to be University

TEAM MEMBERS

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| Academic Year: 2026–27

Introduction & Problem Statement

Background

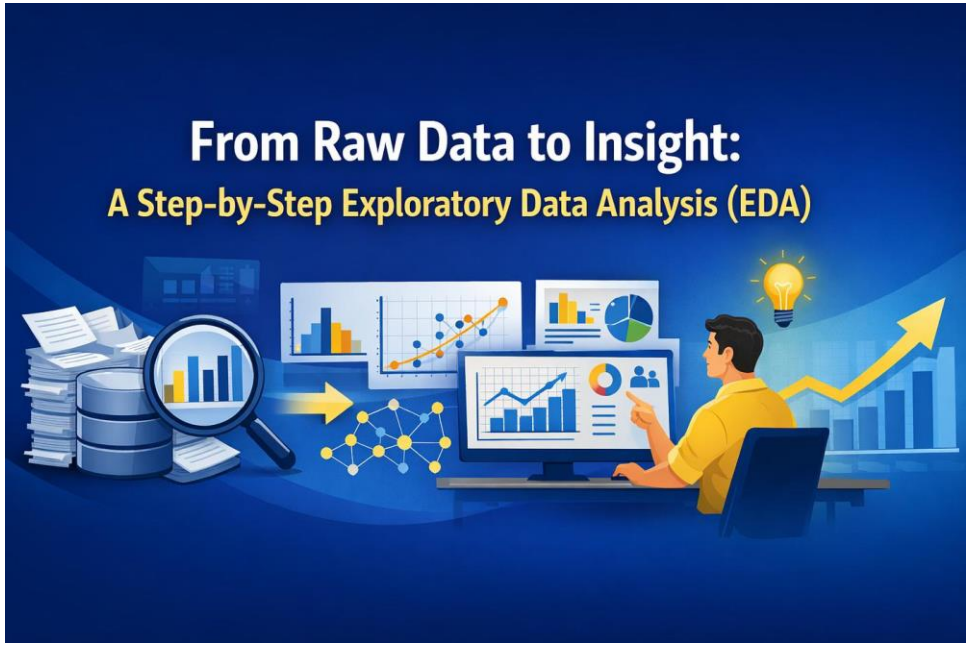
Exploratory Data Analysis (EDA) helps us understand data before performing further analysis. It helps identify patterns, trends, relationships and important information from datasets.

Problem Statement

Analyzing large datasets manually is time-consuming. Users need to write separate code to check data quality, calculate statistics and generate visualizations.

Motivation / Need

[Our project provides a single interactive dashboard where users can upload datasets, analyze data, view statistics and generate visualizations easily without writing separate analysis code.



Objectives & Scope

Objectives

1. Develop an interactive dashboard to upload and analyze different datasets.
2. Analyze missing values, duplicate records and data quality.
3. Generate statistical summaries and interactive visualizations.
4. Identify data patterns, correlations and potential outliers

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Anderson & Jafar (2009)	Visual Exploratory Data Analysis	Interactive visualization helps users identify patterns and relationships in data.	Mainly focuses on visual data exploration.
2	Swayne & Buja (1998)	Interactive High-Dimensional Data Visualization	Supports exploring relationships between multiple variables and missing data.	Limited focus on missing-data exploration.
3	Cheng, Cook & Hofmann (2015)	GUI-Based Missing Value Visualization	Provides graphical and statistical summaries to understand missing values.	Mainly focuses on missing-value analysis.
4	Behringer et al. (2024)	Interactive Data Quality Monitoring	Combines data-quality monitoring with exploratory data analysis.	Focuses mainly on data-quality monitoring.

Research Gap

Existing approaches often focus on individual tasks such as data visualization, missing-value analysis or data-quality monitoring. Our project combines dataset profiling, data-quality checks, statistical analysis, interactive visualizations and insights in one user-friendly dashboard.

Proposed System & Methodology

STEP 1

Define Problem

Identify the need for a simple dashboard to analyze datasets and understand data quality.

STEP 2

Collect Data

Upload datasets in CSV, Excel, JSON and other supported formats for analysis.

STEP 3

Design

Design dashboard layout using Python, Dash and Plotly.

STEP 4

Implement

Develop data analysis, quality checks and interactive charts

STEP 5

Test & Evaluate

Test with sample datasets and verify the results.

Key Novelty / Contribution

Our project combines dataset profiling, data-quality analysis, statistical summaries and interactive visualizations in a single dashboard. It helps users explore datasets and understand important patterns without writing separate analysis code.

System Architecture



Components

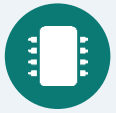
1. Data Input
Upload datasets in supported formats

2. Data Processing
Load and inspect data using Pandas.

3. EDA Analysis
Calculate statistics and check data quality.

4. Visualization
Display interactive charts and insights using Plotly.

Hardware & Software Requirements



Hardware

Processor: Intel Core i3 or above
RAM: 4 GB minimum
Storage: 500 MB free space
Internet connection



Software & Tools

Operating System: Windows 10/11
Programming Language: Python 3.x
IDE: Visual Studio Code
Version Control: Git and GitHub



Libraries / Frameworks

Pandas – Data processing
NumPy – Numerical operations
Dash – Web dashboard framework
Plotly – Interactive visualizations

Deployment / Platform: Local Machine – Web Browser (Localhost)

Dataset / Data Understanding

1,508

Records (rows)

12

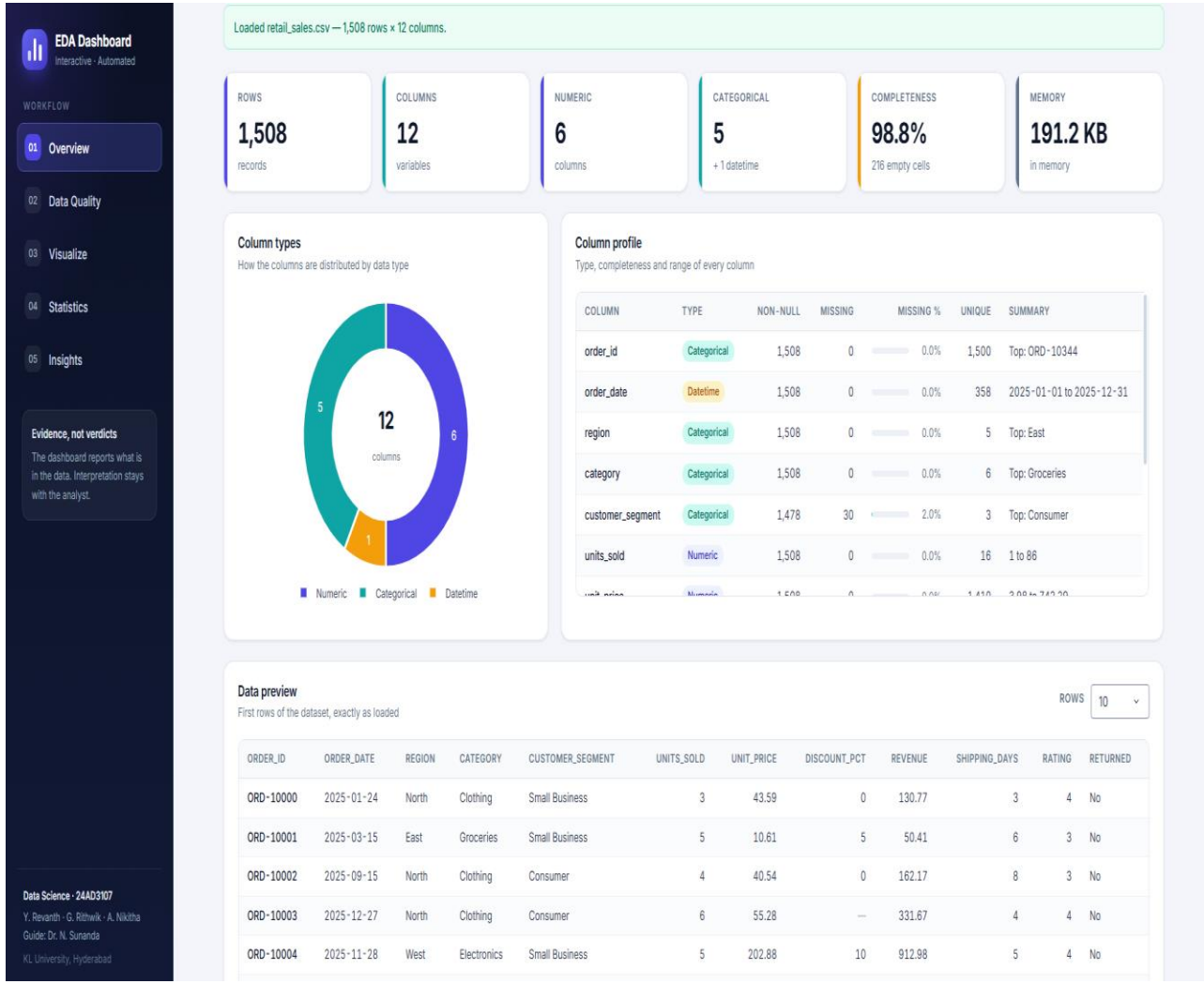
Features (columns)

216

Missing values

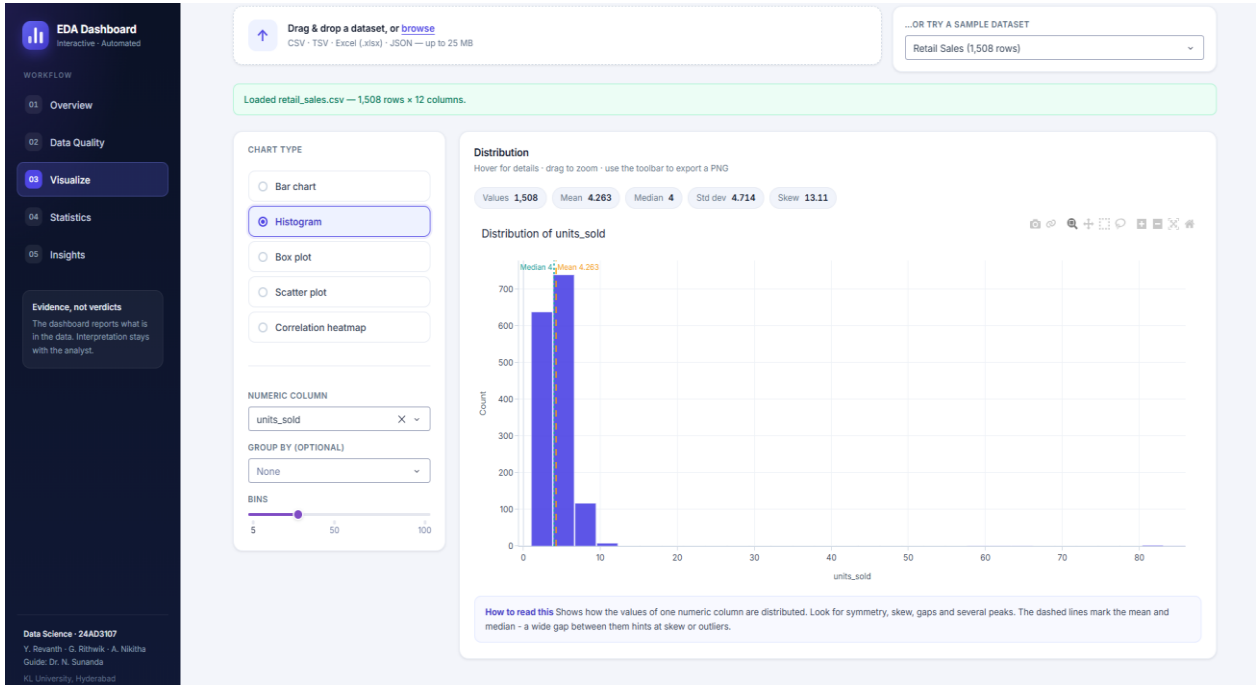
8

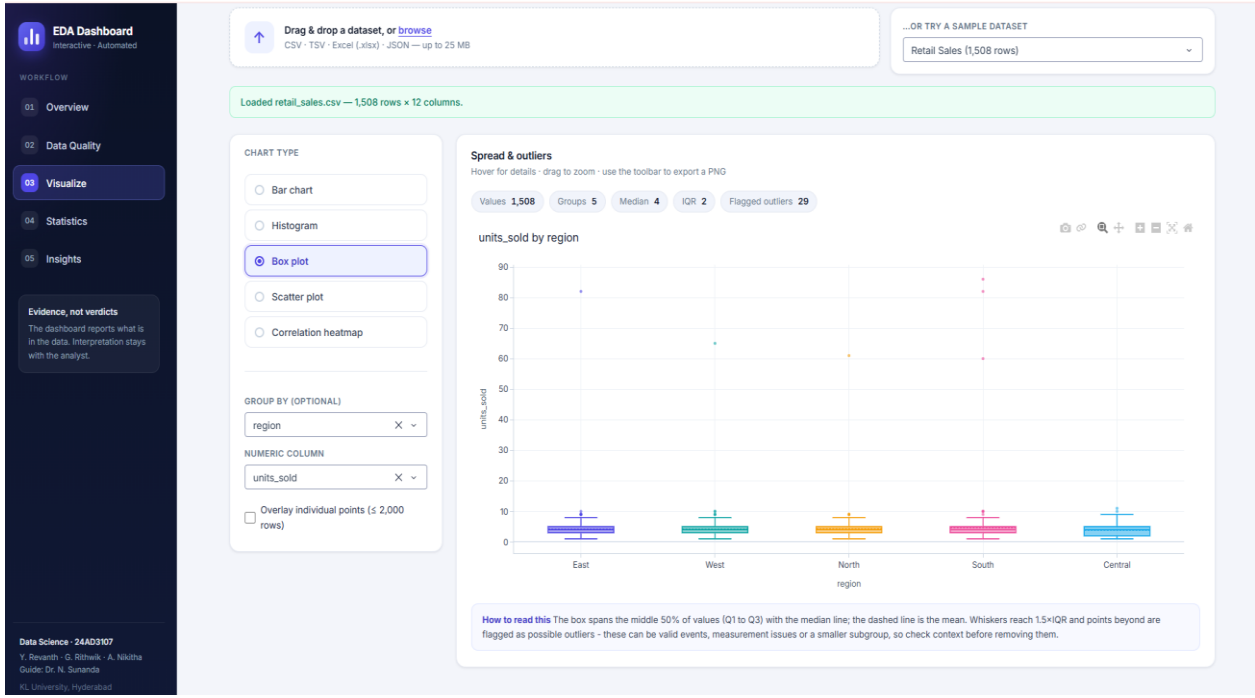
Duplicate Records



- Source: Retail Sales sample dataset (synthetically generated).
- Features: Student-Order ID, date, region, category, customer segment, sales and related attributes.
- Quality: Missing values, duplicate records and data patterns are analyzed.

Data Preprocessing & Exploratory Analysis





Data Quality Analysis

Identifies missing values, duplicate records and text inconsistencies.

Statistical Analysis

Calculates mean, median, standard deviation, quartiles and skewness.

Data Visualization

Generates histograms, bar charts, box plots, scatter plots and heatmaps.

Module-wise Description

1 1. Dataset Overview Module

Purpose: Displays dataset structure, column types and data preview.

Input → Output: Uploaded dataset → Dataset summary and preview.

Developed by: Entire Team

2 2. Data Quality Module

Purpose: Identifies missing values, duplicate records and text issues.

Input → Output: Loaded dataset → Data quality report.

Developed by: Entire Team

3 3. Visualization Module

Purpose: Generates interactive charts to explore data patterns.

Input → Output: Selected columns → Histograms, box plots, scatter plots and heatmaps.

Developed by: Entire Team

4 4. Statistics & Insights Module

Purpose: Calculates statistical measures and presents important data findings.

Input → Output: Statistical summaries and insights.

Developed by: Entire Team

Algorithms & Technologies Used

Python & Pandas

Why chosen: Easy data handling and analysis.

Key parameters: Dataset loading, data profiling, missing-value and duplicate detection.

Dash

Why chosen: Builds an interactive web-based dashboard.
Key parameters: Dashboard layout, user inputs and interactive components

Plotly

Why chosen: Creates interactive and clear data visualizations.
Key parameters: Histograms, bar charts, box plots, scatter plots and correlation heatmaps.

Selection criteria: Selected based on ease of use, interactive visualization, efficient data processing and suitability for exploratory data analysis.

Implementation / Model Building

Feature Selection

Users select numerical and categorical columns for analysis and visualization.

Preprocessing Pipeline

Load datasets using Pandas, inspect data types and analyze missing values and duplicates.]

Analysis & Visualization

Generate statistical summaries, histograms, bar charts, box plots, scatter plots and heatmaps.]

Dashboard Configuration

Provide interactive chart options, column selection, filters and dataset preview.

Code snippet / pseudocode

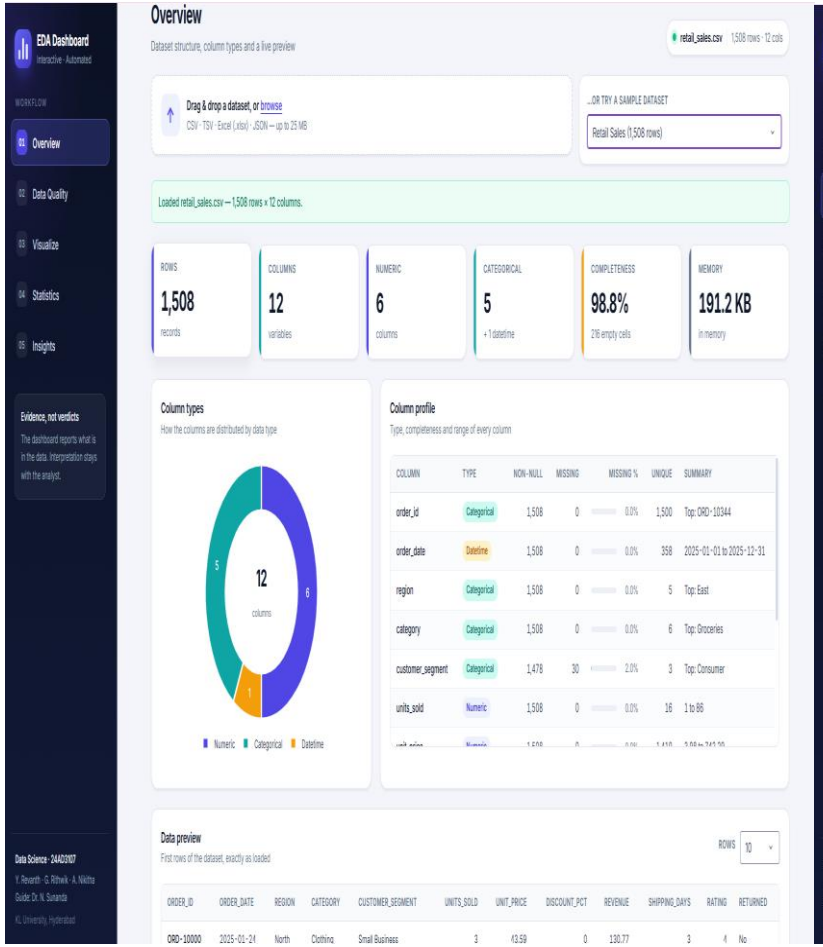
```
# Load dataset
df = pd.read_csv(file_path)

# Analyze dataset
summary = df.describe()
missing = df.isnull().sum()
duplicates = df.duplicated().sum()

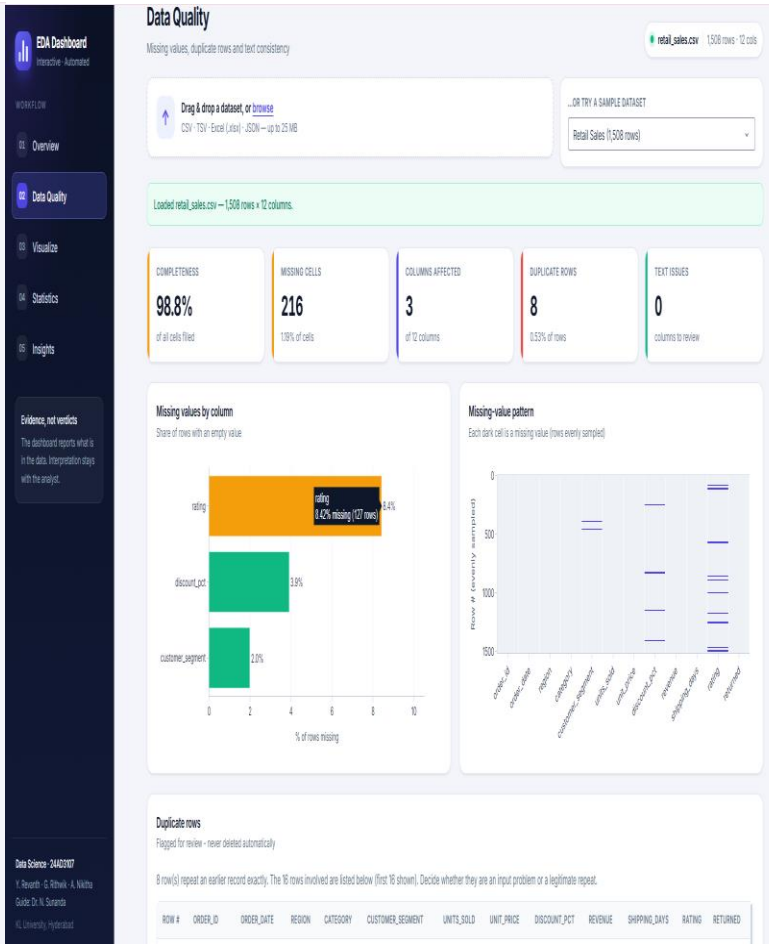
# Generate visualization
fig = px.histogram(df, x="units_sold")

# Display dashboard
app.layout = html.Div([
    dcc.Graph(figure=fig)
])
```

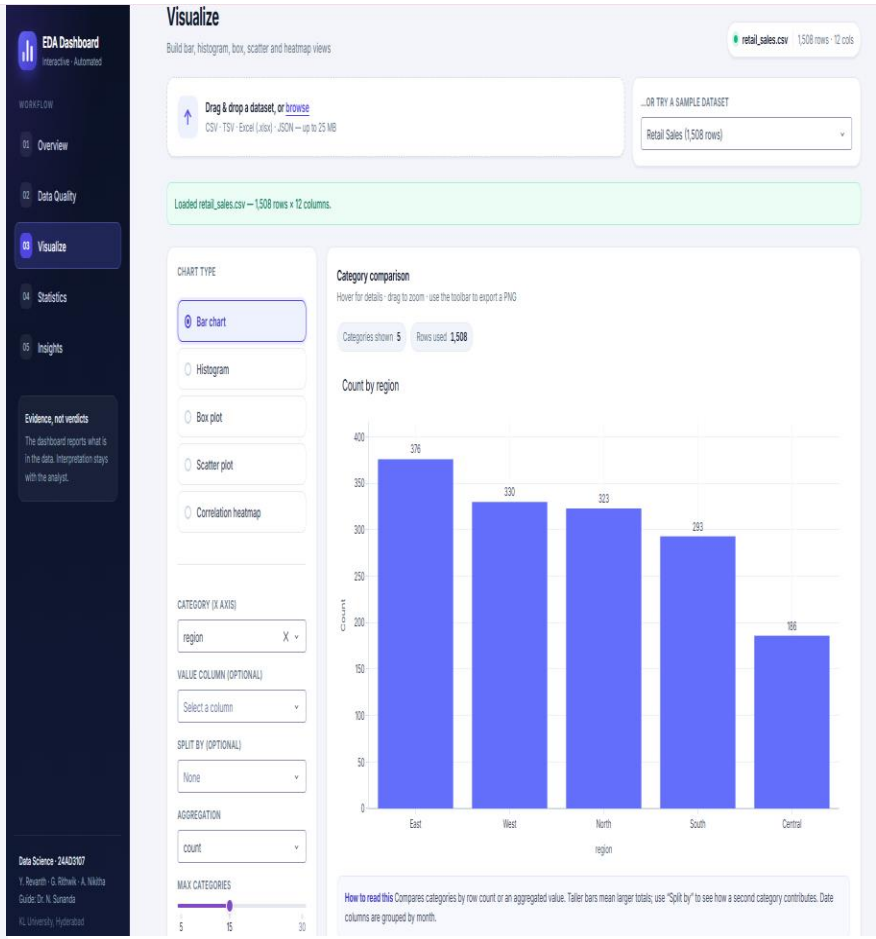
Working Prototype / Output Screens



Overview Dashboard: Displays dataset structure, column types, summary metrics and

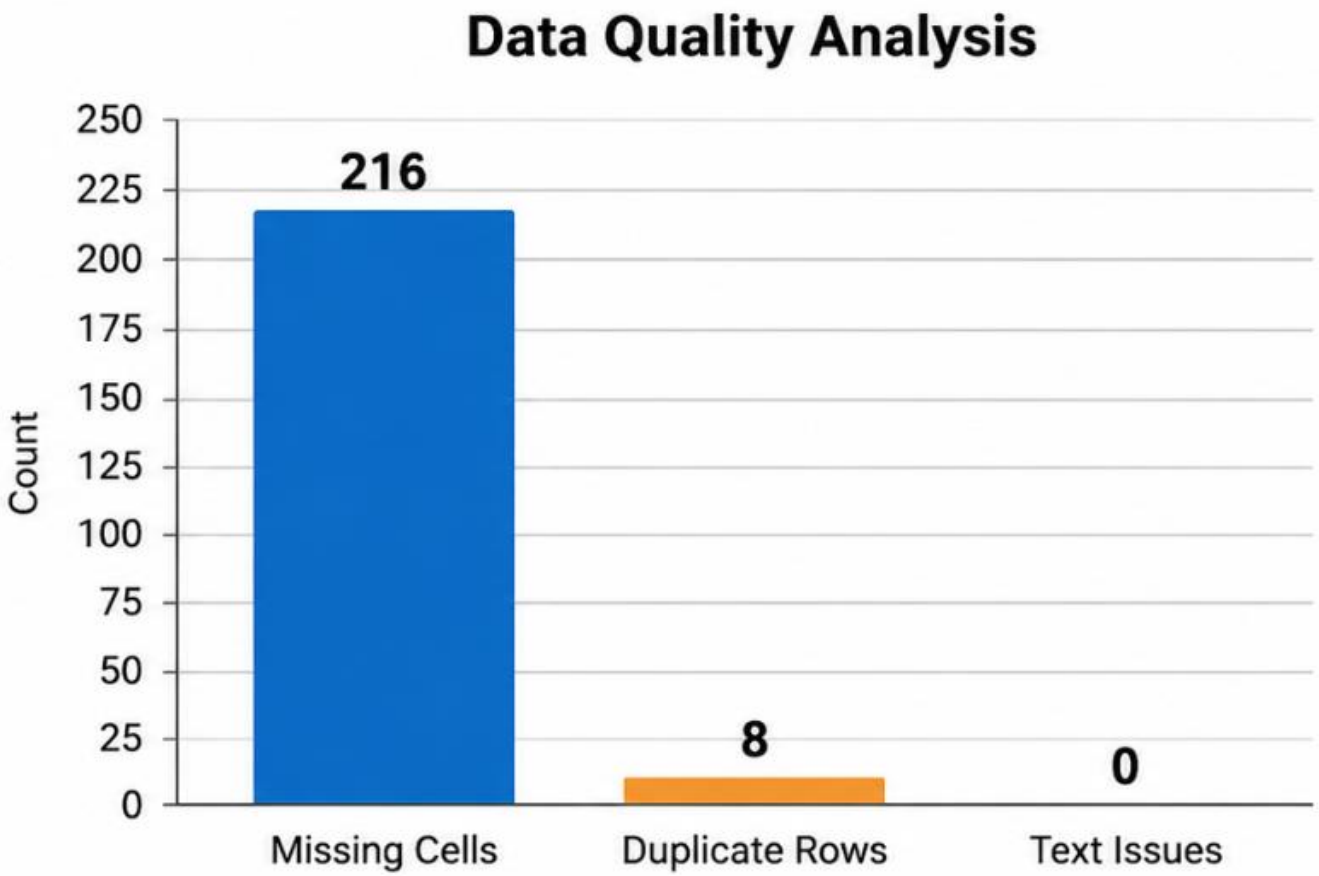


Data Quality Analysis: Identifies missing values, duplicate records and data quality issues.



Interactive Data Visualization: Displays category comparisons across different regions using interactive charts.

Results & Analysis Summary

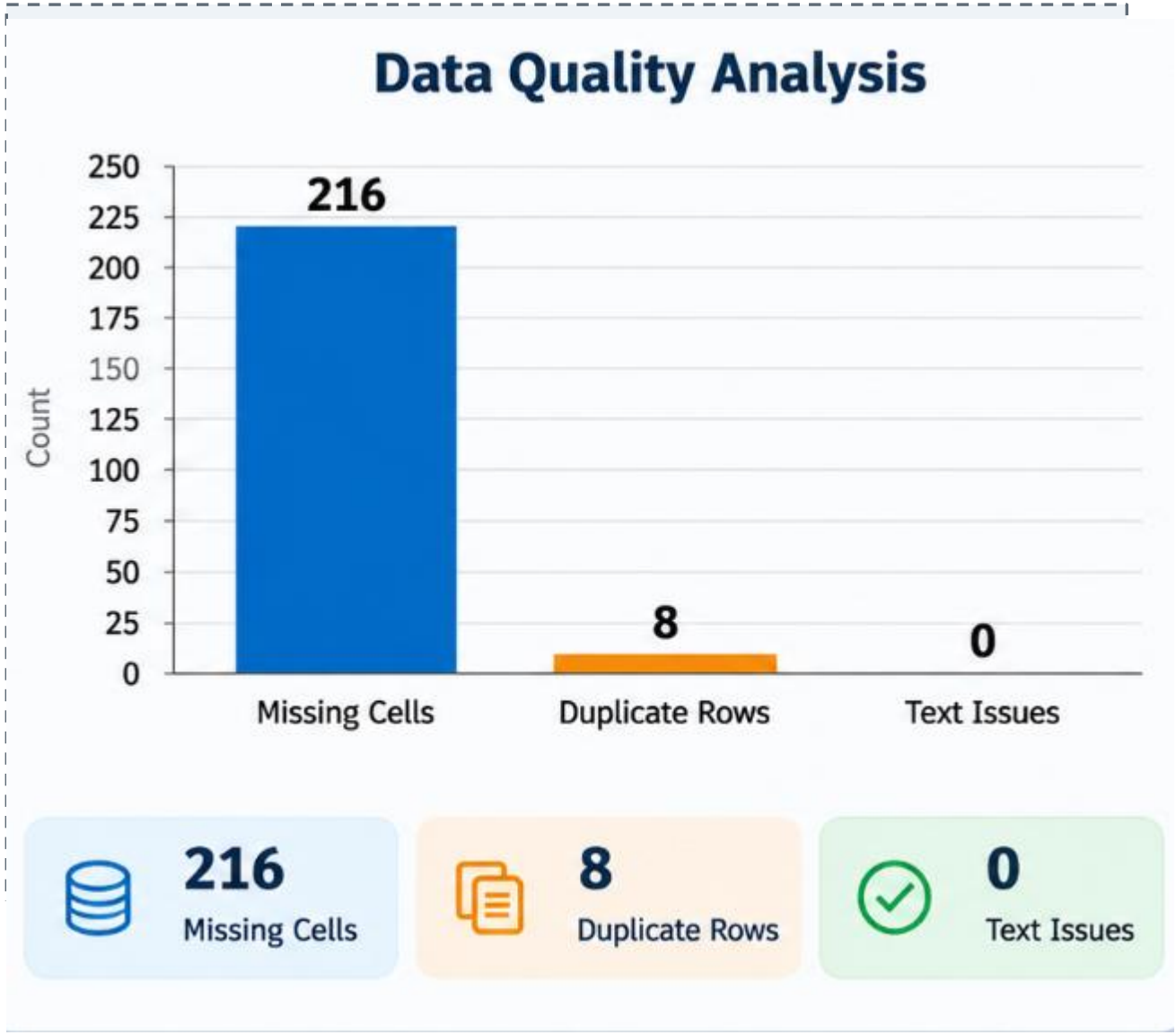


Metric	Result
Total Records	1,508
Total Columns	12
Numeric Columns	6
Categorical Columns	5
Datetime Columns	1
Data Completeness	98.8%

Key Findings: The dashboard successfully analyzes dataset structure, identifies missing values and duplicate records, calculates statistical summaries, and generates interactive visualizations to support data exploration.

Results obtained using the Retail Sales sample dataset.

Analysis & Key Findings



98.8%

Data Completeness

Interpretation: The dataset has 98.8% completeness, indicating that most values are available.
Error analysis: 216 missing cells and 8 duplicate rows were identified.
Key drivers: [Missing values occur in 3 columns. No text consistency issues were detected.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Upload Retail Sales dataset	Dataset loads successfully	1,508 rows and 12 columns displayed	Pass
TC-02	Check missing values	Missing values are identified	216 missing cells detected	Pass
TC-03	Check duplicate records	Duplicate rows are identified	8 duplicate rows detected	Pass
TC-04	Generate visualizations	Selected charts are displayed	Histogram and box plot displayed	Pass

Functional Testing

Dataset upload, data quality checks, statistics and chart generation were tested.

Validation Method

Results were checked using the Retail Sales sample dataset and displayed dashboard outputs.

Performance Testing

Checked dataset loading and dashboard response while switching between analysis sections.

Challenges & Limitations

Challenges Faced

- Handling different file formats → Used Pandas to load supported datasets.
- Identifying data quality issues → Added checks for missing values, duplicate rows and text inconsistencies.
- Visualizing different data types → Used Plotly to generate suitable interactive charts.

Limitations

- File size: Dataset uploads are limited to 25 MB.
- Automatic cleaning: The dashboard identifies issues but does not automatically remove or correct them.
- Insights: Findings are based on statistical analysis and predefined rules, not predictive models.

Conclusion

1

Project Summary
Developed an Interactive EDA Dashboard using Python, Dash and Plotly. It allows users to upload datasets, examine data quality, calculate statistics and generate interactive visualizations.

2

Key Results
The Retail Sales sample dataset contains 1,508 records and 12 columns. The dashboard identified 216 missing cells, 8 duplicate rows and 0 text issues.

3

Project Outcome
The dashboard combines data profiling, quality analysis, statistical summaries and visualizations in one platform, making exploratory data analysis easier and faster.



Final Outcome

98.8%

Data Completeness
Objectives achieved: 4 out of 4

Future Scope & Applications

Improve Performance
Optimize data processing and chart generation for larger datasets.

Expand Data / Features
Add support for more file formats and advanced data analysis features.

Deploy & Build an Interface
Deploy the dashboard online so users can access it from anywhere.

Scale & Secure
Improve data privacy, security and reliability for handling larger datasets.

Applications

Education: Analyze student performance datasets.

Business: Explore sales and customer data.

Research: Identify patterns and relationships in research datasets.

Target Users: Students, data analysts, researchers and business professionals.

References

1. Anderson, R. and Jafar, M. (2009). “A Visual Approach to Exploratory Data Mining.” Review of Business Information Systems, 13(1).

2. Swayne, D. F. and Buja, A. (1998). “Missing Data in Interactive High-Dimensional Data Visualization.” Computational Statistics, 13(1).[3] Organisation, “Dataset or website title,” Year. [Online]. Available: URL

3. 5. Plotly Technologies Inc. “Plotly Python Graphing Library Documentation.” https://plotly.com/python/

THANK YOU

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